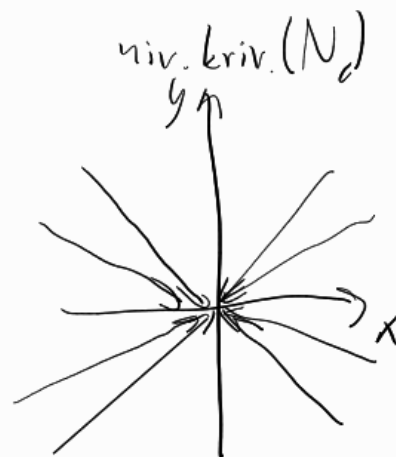
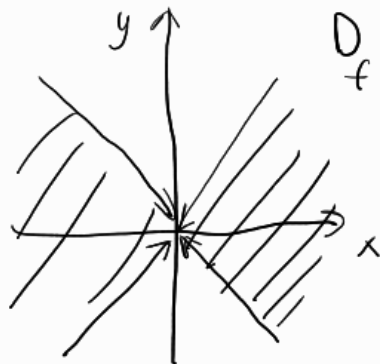


1. Skiciraj D_f , nivojske kriv., graf

a) $f(x, y) = \arcsin \frac{y}{x}$

$$D_f: x \neq 0; \frac{y}{x} \in [-1, 1]$$



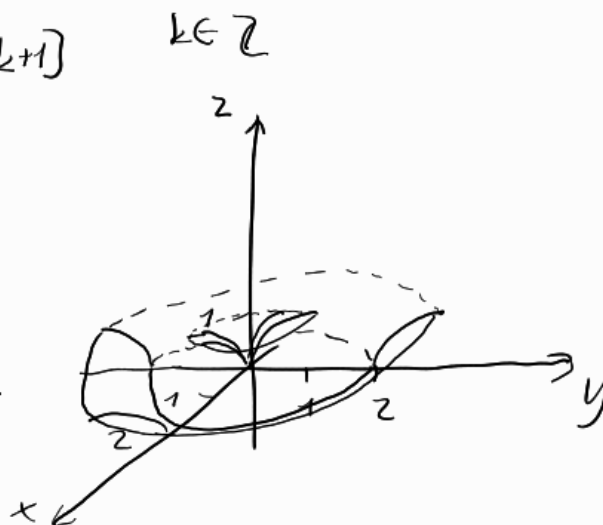
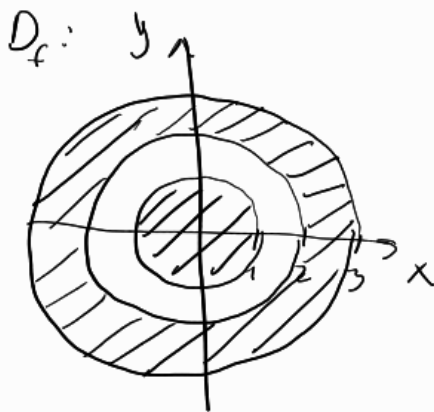
b) $f(x, y) = \sqrt{\sin(\pi \sqrt{x^2 + y^2})}$

$$\sin \pi \cdot a > 0:$$

$$a \in [0, 1] \cup [2, 3] \cup \dots$$

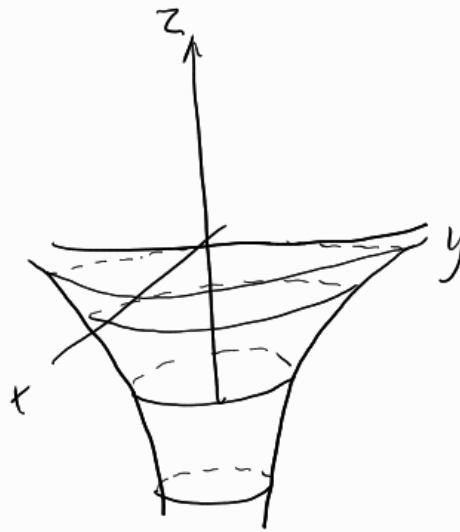
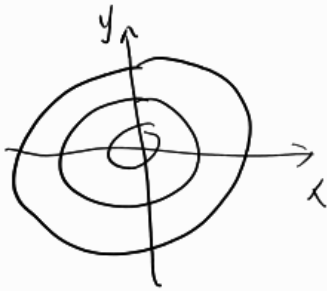
$$a \in [2k, 2k+1]$$

$$\Rightarrow \sqrt{x^2 + y^2} \in [2k, 2k+1] \quad k \in \mathbb{Z}$$



② Nivojnice in gradientno polje potenciala

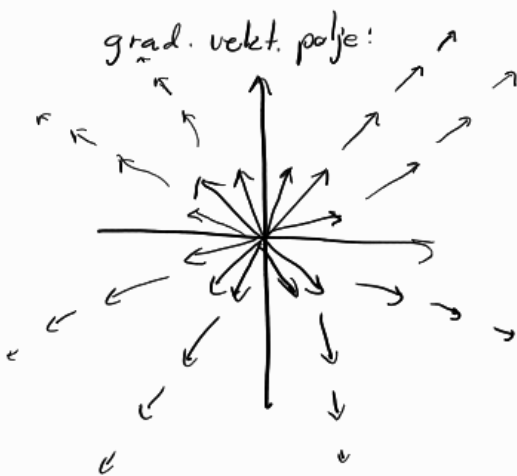
$$V(x, y) = - \frac{mMG}{\sqrt{x^2 + y^2}}$$



$$\nabla V = \left(\frac{mMGx}{\sqrt{(x^2 + y^2)^3}}, \frac{mMGy}{\sqrt{(x^2 + y^2)^3}} \right)$$

$$\sim \frac{\vec{e}_r}{r^2}$$

trajektorija v.p.:



Smerni odvod (v smeri $\vec{s}$): $\vec{s} = (a, b)$

$$D_{\vec{s}} f(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + ah, y_0 + bh) - f(x_0, y_0)}{h} = \nabla f(x_0, y_0) \cdot \vec{s}$$

$$T_{\text{Taylor okoli } (a, b)} f(x, y) = f(a, b) + \left(\frac{\partial f}{\partial x}(a, b)(x-a) + \frac{\partial f}{\partial y}(a, b)(y-b) \right) + \dots + \frac{1}{h!} \sum_{k=0}^h \binom{h}{h-k} \frac{\partial^h f}{\partial x^{h-k} \partial y^k}(a, b) (x-a)^{h-k} (y-b)^k + \dots$$

3. aproksimiraj s Tay. pol. red₃ v (0,0)

$$f(x,y) = (x^2+x+1)\sin y + e^x$$

$$\frac{\partial f}{\partial x} = (2x+1)\sin y + e^x$$

$$\frac{\partial f}{\partial y} = (x^2+x+1)\cos y$$

$$\frac{\partial^2 f}{\partial x^2} = 2\sin y + e^x$$

$$\frac{\partial^2 f}{\partial y^2} = -(x^2+x+1)\sin y$$

$$\frac{\partial^3 f}{\partial x^3} = e^x$$

$$\frac{\partial^3 f}{\partial y^3} = -(x^2+x+1)\cos y$$

$$\frac{\partial^2 f}{\partial x \partial y} = (2x+1)\cos y$$

$$\frac{\partial^3 f}{\partial x^2 \partial y} = 2\cos y$$

$$\frac{\partial^3 f}{\partial x \partial y^2} = -\sin y (2x+1)$$

$$T_3 f(x,y) \approx 1 + (x+y) + \frac{1}{2}(x^2 + 2xy) + \frac{1}{3!}(x^3 + 3 \cdot 2 \cdot x^2 y + 3 \cdot 0 \cdot xy^2 - y^3)$$

$$= 1 + x + y + \frac{x^2}{2} + xy + \frac{x^3}{6} + x^2 y - \frac{y^3}{6} =$$

$$= \left[\frac{x^3}{6} + \frac{x^2}{2} + x^2 y + xy + x + y - \frac{y^3}{6} + 1 \right]$$

2. Način:

$$f(x,y) = (x^2+x+1)\sin y + e^x = (x^2+x+1) \left(y - \frac{y^3}{3!} + \frac{y^5}{5!} - \dots \right) + \left(1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \dots \right)$$

$\underbrace{\quad}_{y - \frac{y^3}{3!} + \frac{y^5}{5!} - \dots} \quad \underbrace{\quad}_{1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\dots}$

$$= x^2 y + xy + y - \frac{y^3}{3!} + 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}$$

← vzamemo tako, da $x^a y^b \rightarrow a+b \leq 3$

$$4. \quad nRT = (V-B)\left(p + \frac{A}{V^2}\right) \rightarrow p = \frac{nRT - \frac{A}{V} + \frac{BA}{V^2}}{V-B}$$

$$T_1 p(T, V) \text{ okoli } (T_0, V_0)$$

$$\frac{\partial}{\partial T}: \quad nR = (V-B) \frac{\partial p}{\partial T} \rightarrow \frac{\partial p}{\partial T} = \frac{nR}{V-B}$$

$$\begin{aligned} \frac{\partial}{\partial V}: \quad 0 &= \left(\frac{\partial p}{\partial V} - 2 \frac{A}{V^3} \right) (V-B) + \left(p + \frac{A}{V^2} \right) = \\ &= \frac{\partial p}{\partial V} (V-B) + p - \frac{A}{V^2} + 2 \frac{AB}{V^3} \rightarrow \frac{\partial p}{\partial V} = \frac{\frac{A}{V^2} - p - 2 \frac{AB}{V^3}}{V-B} = \end{aligned}$$

$$dp = \frac{\partial p}{\partial V} dV + \frac{\partial p}{\partial T} dT$$

$$\begin{aligned} T_1 p(T, V) &= p(T_0, V_0) + \left(-\frac{nRT_0}{(V_0-B)^2} + \frac{2A}{V_0^3} \right) (V-V_0) + \\ &+ \frac{nR}{V_0-B} (T-T_0) \end{aligned}$$

$$\begin{aligned} &= \frac{A}{V^2(V-B)} - \frac{2AB}{V^3(V-B)} - \frac{nRT}{(V-B)^2} + \frac{A}{V(V-B)^2} \\ &- \frac{BA}{V^2(V-B)^2} = \frac{1}{V^3(V-B)^2} \left(AV(V-B) - \right. \\ &- 2AB(V-B) - V^3 nRT + AV^2 - \\ &- ABV \left. \right) = \frac{1}{V^3(V-B)^2} \left(AV^2 - ABV - 2ABV \right. \\ &+ 2AB^2 - V^3 nRT + AV^2 - ABV \\ &= -\frac{nRT}{(V-B)^2} + \frac{2A}{V^3} \end{aligned}$$

Ekstremi funkcij

- stat. točke:

$$\frac{\partial f}{\partial x}(x_0, y_0) = 0 \quad \wedge \quad \frac{\partial f}{\partial y}(x_0, y_0) = 0$$

$$H(x_0, y_0) = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2}(x_0, y_0) & \frac{\partial^2 f}{\partial x \partial y}(x_0, y_0) \\ \frac{\partial^2 f}{\partial x \partial y}(x_0, y_0) & \frac{\partial^2 f}{\partial y^2}(x_0, y_0) \end{bmatrix}$$

$\text{Det } H < 0 \rightarrow \text{sedlo}$

$\text{Det } H = 0 \rightarrow$ ne vemo, gledamo višje odvode

$\text{Det } H > 0$: lok. ekstrem

$$\rightarrow \frac{\partial^2 f}{\partial x^2} > 0 \rightarrow \text{minimum}$$

$$\rightarrow \frac{\partial^2 f}{\partial x^2} < 0 \rightarrow \text{maximum}$$

5. a) $f(x,y) = x^3 + 3xy^2 - 15x - 12y$

$$\frac{\partial f}{\partial x} = 3x^2 + 3y^2 - 15$$

$$\frac{\partial f}{\partial y} = 6xy - 12$$

$$x^2 + y^2 - 5 = 0$$

$$xy - 2 = 0$$

$$x = \frac{2}{y}$$

$$\frac{\partial^2 f}{\partial x^2} = 6x$$

$$\frac{\partial^2 f}{\partial x \partial y} = 6y$$

$$\frac{\partial^2 f}{\partial y^2} = 6x$$

$$\det H = 36x^2 - 36y^2$$

① $T_1(2, 1)$

$$\det H = 36(4-1) > 0$$

$$6 \cdot 2 > 0 \Rightarrow \text{lok. minimum}$$

② $T_2(-1, -2)$

$$\det H = 36(4-1) > 0$$

$$6 \cdot (-2) < 0 \Rightarrow \text{lok. maximum}$$

$$\frac{4}{y^2} + y^2 - 5 = 0$$

$$y^4 - 5y^2 + 4 = 0$$

$$(y^2 - 1)(y^2 - 4)$$

$$y_1 = 1 \quad x_1 = 2$$

$$y_2 = -1 \quad x_2 = -2$$

$$y_3 = 2 \quad x_3 = 1$$

$$y_4 = -2 \quad x_4 = -1$$

③ $T_3(1, 2)$

$$\det H = 36(1-4) < 0$$

$$\Rightarrow \text{saddle}$$

④ $T_4(-1, -2)$

$$\det H = 36(1-4) < 0$$

$$\Rightarrow \text{saddle}$$

b) $f(x,y) = \frac{8}{x} + \frac{x}{y} + y$

$$\frac{\partial f}{\partial x} = -\frac{8}{x^2} + \frac{1}{y}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{16}{x^3}$$

$$\frac{\partial f}{\partial y} = -\frac{x}{y^2} + 1$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{2x}{y^3}$$

$$\frac{\partial^2 f}{\partial x \partial y} = -\frac{1}{y^2}$$

$$-\frac{8}{x^2} + \frac{1}{y} = 0$$

$$-\frac{x}{y^2} + 1 = 0$$

$$x^2 = 8y$$

$$x = y^2$$

$$y^4 = 8y$$

$$y(y^3 - 8) = 0$$

$$\det H = \frac{1}{y} > 0$$

$$y_1 = 0 \quad x_1 = 0 \quad \text{N. Res.}$$

$$y_2 = -2 \quad x_2 = 4$$

$$\frac{1}{y} > 0 \rightarrow \text{lok. min}$$

$$\hookrightarrow f(x, y) = e^{x-y} (x^2 - 2y^2)$$

$$\frac{\partial f}{\partial x} = e^{x-y} ((x^2 - 2y^2) + 2x)$$

$$\frac{\partial f}{\partial x^2} = e^{x-y} (x^2 + 2x - 2y^2 + 2x + 2)$$

$$\frac{\partial f}{\partial y} = e^{x-y} ((-1)(x^2 - 2y^2) - 4y)$$

$$\frac{\partial f}{\partial y^2} = e^{x-y} (x^2 - 2y^2 + 4y + 4y - 4)$$

$$\frac{\partial f}{\partial x \partial y} = e^{x-y} (2y^2 - x^2 - 4y - 2x)$$

$$e^{x-y} (x^2 - 2y^2 + 2x) = 0$$

$$e^{x-y} (2y^2 - 4y - x^2) = 0$$

$$2y^2 - 4y - 2y^2 + 2\sqrt{2y^2 - 4y} = 0 \quad x^2 = 2y^2 - 4y$$

$$2y = \sqrt{2y^2 - 4y}$$

$$y_1 = 0 \quad x_1 = 0$$

$$y_2 = -2 \quad x_2 = \sqrt{8 + 8} = 4$$

$$4y^2 = 2y^2 - 4y$$

$$2y^2 + 4y = 0 \quad y(y+2) = 0$$

$$H = \begin{vmatrix} e^{x-y}(x^2+4x-2y^2+2) & e^{x-y}(2y^2-4y-x^2-2x) \\ e^{x-y}(2y^2-4y-x^2-2x) & e^{x-y}(x^2-2y^2+8y-4) \end{vmatrix}$$

$$\det H = (e^{x-y})^2 \left((x^2+4x-2y^2+2)(x^2-2y^2+8y-4) - (2y^2-4y-x^2-2x)^2 \right)$$

① $T(0,0)$

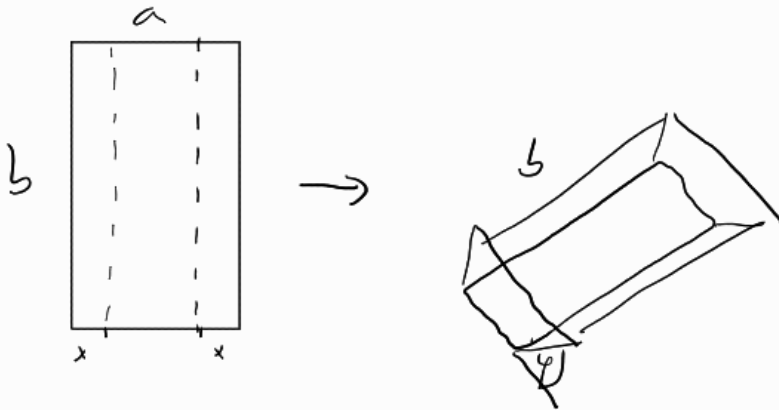
$$\det H = 1(-8) < 0 \Rightarrow \text{sedlo}$$

② $T(4,-2)$

$$\begin{aligned} \det H &= (e^6) \left((16+16-2 \cdot 4+2)(16-2 \cdot 4-16-4) - (2 \cdot 4+8-16-8)^2 \right) = \\ &= e^6 \left((26)(-12) - (64) \right) < 0 \Rightarrow \text{sedlo} \end{aligned}$$

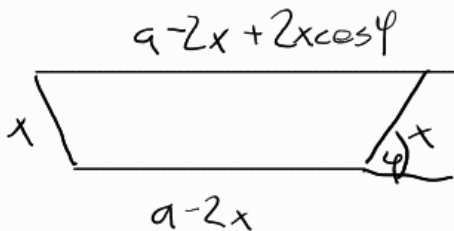
6.

Pravokoten karton s stranicama dolžine a in b prepognemo na razdalji x vzdolž obeh daljših stranic za kot ϕ . Določi x in ϕ tako, da bo volumen dobljenega telesa maksimalen.



$$\phi \in [0, \frac{\pi}{2}]$$

$$x \in [0, \frac{a}{2}]$$



$$V = b \cdot \left(\frac{a-2x+a-2x+2x\cos\phi}{2} \cdot x\sin\phi \right)$$

$$= b \left((a-2x+x\cos\phi)x\sin\phi \right) = h(x, \phi)$$

$$\frac{\partial V}{\partial x} = b \sin\phi \left((-2 + \cos\phi) \cdot x + (a-2x+x\cos\phi) \cdot 1 \right) = b \sin\phi (-4x + \cos\phi x + a)$$

$$\frac{\partial V}{\partial \phi} = b \left(x \cos\phi (a-2x+x\cos\phi) + x \sin\phi (-x \sin\phi) \right)$$

$$\sin \varphi (-4x + \cos \varphi (1+x) + a) = 0 \quad x \cos \varphi (a - 2x + x \cos \varphi) - x^2 \sin^2 \varphi = 0$$

$$a \sin \varphi - 4x \sin \varphi + (1+x) \sin \varphi \cos \varphi = 0 \quad x^2 \cos^2 \varphi - x^2 \sin^2 \varphi + x a \cos \varphi - 2x^2 \cos \varphi = 0$$

$\uparrow$
 $1 - \cos^2 x$

$\sin \varphi \neq 0$

$$a - 4x + 2x \cos \varphi = 0$$

$$\cos \varphi = \frac{4x - a}{2x}$$

$$x^2 \left(2 \left(\frac{4x - a}{2x} \right)^2 - 1 \right) + x a \frac{4x - a}{2x} - 2x^2 \frac{4x - a}{2x} = 0$$

$$x^2 \left(\frac{16x^2 - 8xa + a^2}{2x^2} - 1 \right) + 2ax - \frac{a^2}{2} - 4x^2 + ax = 0$$

$$8x^2 - 4xa + \frac{a^2}{2} - x^2 + 2ax - \frac{a^2}{2} - 4x^2 + ax = 0$$

$$3x^2 - ax = 0$$

$$x_1 = 0 \quad \text{Ni res.}$$

$$x_2 = \frac{a}{3} \quad \cos \varphi = \frac{\frac{4}{3}a - a}{\frac{2}{3}a} = \frac{\frac{1}{3}a}{\frac{2}{3}a} = \frac{1}{2}$$

$$\boxed{\varphi_2 = \frac{\pi}{3}}$$

$$V_{\max} = b \left(\left(a - \frac{2}{3}a + \frac{1}{3}a \cdot \frac{1}{2} \right) \cdot \frac{a}{3} \cdot \frac{\sqrt{3}}{2} \right) =$$

$$= b \left(\left(\frac{1}{2}a \right) \frac{a\sqrt{3}}{6} \right) = \underline{\underline{b \cdot \frac{a^2 \sqrt{3}}{12}}}$$

$$V = b \left((a - 2x + x \cos \varphi) x \sin \varphi \right)$$

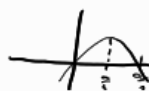
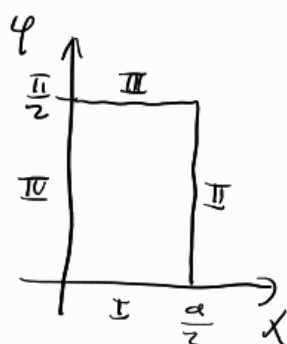
$$V_{\text{I}}(\varphi = 0) = 0$$

$$V_{\text{II max}} = b \frac{a^2}{8}$$

$$V_{\text{II}}(x = 0) = 0$$

$$V_{\text{II}}(x = \frac{a}{2}) = b \left(\frac{a}{2} \cos \varphi \cdot \frac{a}{2} \sin \varphi \right) = b \cdot \frac{a^2}{4} \sin \varphi \cos \varphi = b \frac{a^2}{8} \sin 2\varphi$$

$$V_{\text{II}}(\varphi = \frac{\pi}{2}) = b(a - 2x)x \rightarrow \max \text{ pri } x = \frac{a}{2} \rightarrow V_{\text{II max}} = b \left(\frac{a}{2} \right) \frac{a}{2} = \underline{\underline{b \frac{a^2}{8}}}$$



$$\frac{ba^2}{8} \text{ vs. } \frac{ba^2}{8} \text{ vs. } ba^2 \frac{\sqrt{3}}{12}$$

$$\frac{1}{8} \text{ vs } \frac{\sqrt{3}}{12} \Rightarrow V_{\max}\left(\frac{a}{3}, \frac{\pi}{3}\right) = \frac{ba^2\sqrt{3}}{12}$$